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Evaluating Performance Enhancement of Traditional Machine Learning Ensembles for Cross-Source Phishing Email Detection on the MeAJOR Dataset

Abstract

Phishing is one of the costliest types of cyberattacks, with the average cost of a data breach equaling more than \$4.88 million per breach (IBM, 2024). While much research has been conducted, most phishing email detection systems to date use uniform, single-source corpora that overestimate performance metrics when tested in the lab and underperform in the wild. This research designs and builds a full-stack machine learning pipeline to detect phishing emails using a 103,340-sample corpus of multi-source emails from the MeAJOR dataset (Mendes, Maia and Praça, 2025). It combines NLP preprocessing with lemmatisation (via the NLTK WordNetLemmatizer) and TF-IDF feature engineering using unigrams, bigrams and trigrams with a 20,000-feature vocabulary, stratified 60/20/20 training/validation/test set splits, and six individually hyperparameter-optimised classifiers (Logistic Regression, LinearSVC, Multinomial Naive Bayes, Random Forest, AdaBoost, k-Nearest Neighbours) combined via hard and soft voting ensemble methods. The Soft Voting ensemble delivers 97.87 percent macro F1 on the test set and a mean Leave-One-Source-Out (LOSO) F1 of 84.94 (across three source folds) with McNemar's test confirming statistical significance ($p < 0.0001$) over all individual classifiers. The pipeline's transparency is guaranteed by analysing the Logistic Regression coefficients, which reveal the discriminative vocabulary learned by the model. The 12.93 percentage point difference between macro F1 and LOSO F1 measures the cross-source generalisation problem and highlights the need for multi-source evaluation frameworks in the study of phishing detection.

Keywords: Phishing, ensemble, TF-IDF, NLP, generalisation, cybersecurity, LOSO

Introduction:

Phishing remains a persistent and costly cybersecurity threat, driven by deceptive communications that impersonate legitimate organisations and exploit human behaviour. It accounts for approximately 15% of global data breach vectors, with an average cost of \$4.88 million per incident (IBM, 2024). The scale of the problem continues to grow: the Anti-Phishing Working Group (APWG) reported over 1.3 million unique phishing websites in 2021, representing a more than 200% increase since 2019 (Bountakas and Xenakis, 2023; Basit et al., 2020). Traditional detection mechanisms such as rule-based and blacklist systems struggle to keep pace with increasingly sophisticated adversaries leveraging domain generation algorithms, fast-flux infrastructures, and advanced social engineering techniques (Henry, Stephen and Asuquo, 2026).

Machine learning has emerged as a promising approach for phishing detection, with classifiers such as Logistic Regression, Random Forest, Support Vector Machines, and XGBoost demonstrating strong performance in controlled evaluations (Rehman et al., 2023; Jeswin et al., 2025). More recently, ensemble methods particularly those based on voting strategies have shown potential for improving predictive stability and reducing variance by combining multiple base learners (Bountakas and Xenakis, 2023; Anirudh S et al., 2024). However, two critical limitations remain. First, most studies evaluate models on single-source datasets, resulting in overfitting to dataset-specific linguistic patterns and reduced generalisation capability. Empirical evidence shows that cross-dataset transfer can lead to performance drops of 30–40%. Second, inconsistencies in dataset quality including class imbalance, outdated samples, and differing preprocessing pipelines hinder fair comparison and reproducibility (Mendes, Maia and Praça, 2025).

This research addresses these challenges by investigating the effectiveness of a multi-source learning approach using the heterogeneous MeAJOR dataset.

Research question: Does a majority-voting ensemble of diverse traditional machine learning classifiers significantly improve classification robustness and cross-source generalization performance compared to individual models in phishing email detection?

To answer this, the study constructs a principled ensemble pipeline based on text-only TF-IDF representations trained across multiple data sources. It evaluates performance both in-distribution and under Leave-One-Source-Out (LOSO) conditions to assess generalisation to unseen domains. Additionally, it analyses whether majority-voting ensembles yield statistically significant improvements over individual classifiers. The remainder of this report presents related work (Section 2), methodology (Section 3), quantitative results (Section 4), model interpretability (Section 5), and limitations with directions for future research (Section 6).

2. Related Work

2.1 Machine Learning-based Phishing Detection

Machine learning methods have been widely used for phishing email detection, with two major paradigms of features commonly used in the literature: content-based features - extracted from email structural information including headers, URLs and attachments; and text-based features - extracted from the body of the email using natural language processing (NLP) techniques such as TF-IDF vectorisation and word embeddings (Bountakas and Xenakis, 2023). TF-IDF has proven to be a successful approach, allowing models to recognise suspicious linguistic patterns while filtering out uninformative terms. Jeswin et al. (2025) showed this with 98.60 percent accuracy on five classifiers (majority-vote ensemble on TF-IDF features), and 99.04 percent from SVM alone. But Jeswin et al. (2025) test on a single amalgamated dataset without source-specific splits, and so it is not possible to tell if the model has generalised or overfit to the source - a common issue addressed by the LOSO protocol in this project.

Kumar S et al. (2025) augmented TF-IDF features with contextual information - reputation for the email's sender, and temporal patterns of email activity - to reach 0.917 accuracy and F1 of 0.87 on the Enron Corpus. Although contextual enrichment helped minimise false positives under adversarial attacks, it only experimented with 200 aggregated records, limiting the ability to infer model generalisation on a larger corpus. A crucial datasets contribution is that of Mendes, Maia and Praça (2025), whose MeAJOR Corpus brings together 135,894 labelled emails from five open-source sources, with uniform processing and multiple category feature engineering - attaining 98.34 percent F1 using XGBoost on text and URL features. While MeAJOR significantly improves corpus quality, they do not use source-aware splits, and do not address cross-source generalisation - the focus of this project.

2.2 Ensemble Learning Methods

Ensemble methods use multiple classifiers to make more consistent predictions by blending multiple decision boundaries. Ensemble methods typically perform better than individual weak classifiers, although they may not improve the performance of the best individual classifier, and may even perform worse in certain measures. Kalaiselvi et al. (2024) integrated Bat Algorithm, Random Forest and SVM in a Knowledge Discovery model, tested on 8,208 emails, with 57 features. Bat Algorithm had the best individual AUC (99.5 percent) and the ensemble (FML) had an AUC of 98.6 percent and F-measure of 92.4 percent. This highlights a key point: the ensemble performs well across multiple metrics, and is less reliant on any one learner, rather than surpassing the best-performing individual learner in all metrics. An ensemble strategy (Stacking) by Anirudh S et al. (2024) - using SVM and XGBoost as base learners and Logistic Regression as a meta-learner - achieved 97.92 percent accuracy and F1 of 98.32 percent on 18,650 emails, outperforming all individual models, although by less than one percentage point compared to Random Forest at 97.06 percent, indicating that ensemble performance gains on a high-performing system are marginal.

The most comprehensive ensemble study for phishing email is HELPHED (Bountakas and Xenakis, 2023) that uses hybrid content-based and text-based features with parallel Stacking and Soft Voting evaluated on a realistically imbalanced set of 35,511 emails. Soft Voting's F1 of 0.9942, MCC of 0.967 and training time of 0.0313 seconds represents the new state-of-the-art. Crucially, HELPHED's improved performance is contingent on hybrid features that combine both the structure and content of emails, a feature set that was not used in previous text-only approaches. This suggests the hybrid feature matrix as a future work goal of this study. This was confirmed by Rehman et al. (2023) and Basit et al. (2020) on the UCI, suggesting that ensemble aggregation is a safe approach.

2.3 Cross-Dataset Generalisation Challenges

The almost universal practice in the field of phishing detection of testing on single datasets generates over-inflated performance measures that mispredict deployment performance. The most

comprehensive analysis of this issue was provided by Henry et al. (2026): without domain adaptation, the transfer of a Random Forest model between the UCI Phishing Dataset and contemporary PhiUSIIL dataset (which share no common features) resulted in a 42.6-percentage-point drop in accuracy, to 57.4 percent. Their Cross-Domain Ensemble Probability Fusion (CDEPF) approach restored performance to 94.4 percent by harmonising the data distributions through Principal Components Analysis (PCA), adapting domain differences through CORAL and optimising weights on the validation dataset, with strong statistical significance ($t = 98.78$, $p < 0.001$, Cohen's $d = 42.20$). Mia, Derakhshan and Pritom (2024) have brought this insight further using SHAP explainability with two phishing URL datasets, revealing that while the same features are used in each dataset, they contribute differently to phishing detection when training and test datasets are drawn from different sources; direct cross-dataset testing reduced accuracy to 51 percent, while training on a merged multi-source dataset restored accuracy to 91 percent. Although Mia et al. (2024) specifically address the importance of URL features rather than email text tokens, their general insight that feature importance rankings vary across datasets (even with overlapping feature sets), suggests that caution should be exercised when generalising any feature-based phishing classifier, including the TF-IDF email representations used here. Together, these studies demonstrate that cross-source evaluation is a prerequisite for reliable phishing detection studies, in this project achieved via LOSO cross-validation.

2.4 Interpretability and Research Gaps

Models need to be understandable as well as effective if they are to be implemented in practice for cybersecurity. Mia, Derakhshan and Pritom (2024) show how SHAP analysis reveals data-dependent feature behaviour to provide a test for model validation prior to deployment. The analysis of Logistic Regression coefficients is a supplementary, low-cost approach to transparency that allows inspection of the specific terms driving classification - an approach taken in this project in accordance with explainability considerations recognised elsewhere (Anirudh S et al., 2024; Bountakas and Xenakis, 2023).

There are two well-recognised research gaps that motivate this work. Firstly, no cross-source evaluation of email-based phishing detection systems exists: most top-performing systems, such as HELPHED (Bountakas and Xenakis, 2023) and the stacking model of Anirudh S et al. (2024), are evaluated solely on single held-out folds of the training distribution. This makes it impossible to evaluate whether learned representations are stable and generalise to other email from different sources, such as campaigns, organisations or time. The second is the lack of hybrid text and structural feature pipelines thoroughly assessed in a cross-source setting. Although MeAJOR (Mendes, Maia and Praça, 2025) and HELPHED both showcase the performance improvement that can be achieved by hybrid feature pipelines, neither assesses cross-source stability by means of a source-aware protocol like LOSO. This research fills the first gap, and provides the empirical grounds for the second one with its Loso protocol and the hybrid features extension described in Section 6.

3. Methodology and Implementation

In this section, we describe the end-to-end machine learning pipeline for detecting phishing emails. The pipeline consists of six interconnected steps in which the data are: collected and analysed, preprocessed using NLP techniques, transformed into TF-IDF features, split into training and test sets, classified, combined into an ensemble, and tested for generalization. The design choices are informed by the literature and made with regard to the two goals of high accuracy and generalisation.

3.1 Dataset Overview and Construction

We use the MeAJOR (Merged email Assets from Joint Open-source Repositories) Corpus, released by Mendes, Maia and Praça (2025) as a principled multi-source dataset for phishing email detection, as the basis for our project. The MeAJOR Corpus is a merged corpus of 135,894 labelled emails from five open-source databases: the Nazario Phishing Corpus, Nigerian Fraud letters, TREC competitive data, among others. The version used in this project is the preprocessed CSV version (available on Zenodo - record 18471483) with 108,685 emails following the dataset authors' own cleaning and deduplication process. After a further deduplication step based on tokenised matches of email text, the final modelling corpus is comprised of 103,340 emails (108,685 raw emails minus 5,344 duplicates). It is important to use a multi-source corpus because models trained on single-source corpora overfit to source-specific features (see the LOSO protocol in Section 3.7).

3.1.1 Exploratory Data Analysis

Before modelling, an exploratory data analysis was carried out to describe the data. The 103,340-sample corpus is comprised of 57,088 benign emails (55.2 percent) and 46,252 phishing emails (44.8 percent) - a small but significant class imbalance that is addressed through class-weighting and oversampling techniques in Section 3.5. The distribution of samples across sources is heterogeneous: TREC-05 is the most balanced and contributes the largest number of samples of both classes, TREC-07 has a greater proportion of phishing samples than benign, while the proportion of phishing samples in TREC-06 is the smallest - a distributional skew that affects the LOSO fold results as described in Section 4.5.

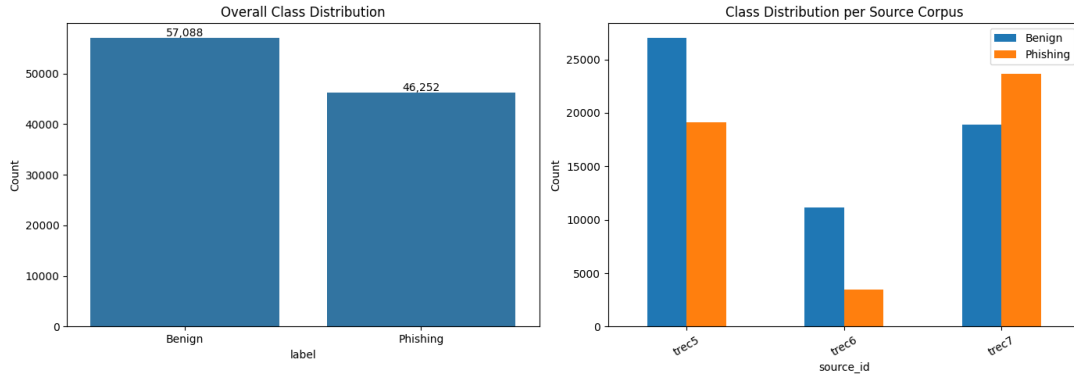


Figure 1: Class Distribution + Per-Source Distribution

The lengths of the processed email bodies are right skewed with a median of around 120 tokens, and a significant tail of lengthy commercial emails (over 2,000 tokens). The count of URLs per email is also right-skewed: most benign emails contain zero URLs, whereas the mean URL count of phishing emails is significantly higher (confirming the high discriminative value of the "url" TF-IDF feature in Section 5.1).

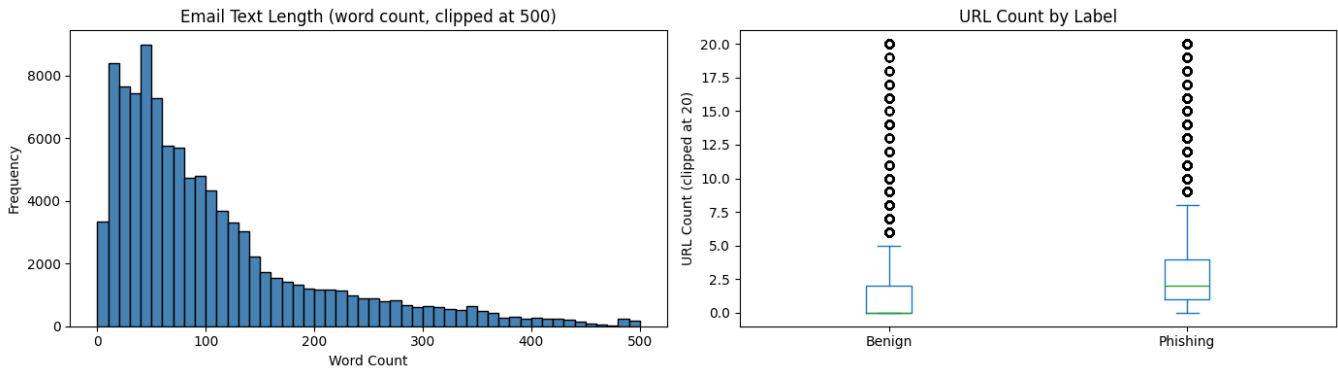


Figure 2: Email Text Length + URL Count by Label

3.2 NLP Preprocessing Pipeline

All emails are preprocessed with a standard five-step NLP pipeline: lowercasing, removal of punctuation and special characters, tokenization, stop word removal and lemmatization. Lowercasing ensures all characters are in the same case to avoid considering the same underlying character as different features. Punctuation removal removes HTML tags, URLs and other artefacts that are common in phishing emails, but lack broad semantic significance. Tokenization breaks the preprocessed text into words, the building blocks for n-grams. Stop word removal eliminates

common but semantically empty words (articles, conjunctions, common prepositions) that would form the biggest component of TF-IDF feature vectors and obliterate discriminative phishing words, as advocated by Anirudh S et al. (2024).

We lemmatised all preprocessed texts using the NLTK WordNetLemmatizer, using Part-of-Speech (POS) tagging before lemmatisation for better morphological accuracy. This transforms inflected word forms to their morphological base (e.g., "verifying" and "verified" are both lemmatised to "verify"), reduces vocabulary sparsity and enhances n-gram consistency. Bountakas and Xenakis (2023) used WordNet lemmatisation in the HELPHED pipeline and noted an improvement in the quality of text-based features without loss of semantic information. The preprocessing pipeline is implemented in scikit-learn's text utilities and NLTK, and is fully reproducible. All random pipeline operations are initialised with a fixed global random seed and all models as well as the TF-IDF vectoriser are serialised with Python's joblib library for reuse.

3.3 Vectorisation: TF-IDF

The text from preprocessed emails is represented as feature vectors using Term Frequency-Inverse Document Frequency (TF-IDF) vectorisation. TF-IDF places greater emphasis on terms that are frequent in a specific email, but rare in the entire email collection, to distinguish terms of discriminative nature for different email classes (Jeswin et al., 2025). This is especially important in phishing detection, where words like "urgent", "verify" and "click here" are highly discriminative of phishing emails but infrequent in legitimate emails (Mendes, Maia and Praça, 2025).

A maximum of 20,000 features is used in the vectoriser. This is typical for text classification problems of a similar scale (Jeswin et al., 2025), for which vocabularies of more than 20,000 features typically provide only marginal gains in discrimination, but greatly increase memory consumption and training time. The n-gram extractor is set to extract unigrams, bigrams and trigrams (`ngram_range = (1, 3)`). Unigrams represent single words; bigrams represent two words context phrases (e.g. "login credentials" or "verify account" etc.) and trigrams represent three words phishing commands (e.g. "click the link") etc. Scikit-learn's sparse matrix format is maintained to ensure computational efficiency of the six different classifiers.

3.4 Data Splitting Strategy

The vectorised dataset is divided into three mutually exclusive subsets in a 60/20/20 ratio by stratified sampling for training, validation and held-out testing, respectively. Stratification ensures the training, validation and test sets reflect the class proportions of phishing and benign emails, crucial to avoid bias in the evaluation of class imbalance (Kumar S et al., 2025). The validation subset can be used for parameter selection and ensemble modelling. Hyperparameter tuning is done using RandomizedSearchCV, with three-fold cross-validation on the training subset only to ensure that the validation and test subsets are not used for hyperparameter tuning. The test set is not touched until final testing. A random seed is set for all trains-test-splits.

3.5 Selection of Classifiers, Class Imbalance and Hyperparameter Tuning

The TF-IDF matrix is used to train six diverse classifiers from multiple families: Logistic Regression, Multinomial Naive Bayes (MNB), LinearSVC, Random Forest, AdaBoost and k-Nearest Neighbours (kNN). The diversity of classifiers is strategic: models with different inductive biases exhibit uncorrelated classification errors, and thus can be combined using voting to reduce classification variance (Bountakas and Xenakis, 2023; Rehman et al., 2023).

Two approaches are taken to address class imbalance. For Logistic Regression, LinearSVC and Random Forest, the `class_weight='balanced'` flag is set, which automatically resamples the loss function to upweight the minority phishing class - that is, to increase the penalty for erroneous classification of the phishing class. For Multinomial Naive Bayes and AdaBoost, which do not support reweighting of the loss function in the same way, a `RandomOverSampler` (ROS) is added to the scikit-learn pipeline, before the classifier is fitted. ROS is applied on each fold of the training data to avoid data leakage. The standard scikit-learn implementation of kNN does not support class weighting or oversampling (hence the comparatively low FN in Section 4.2).

Hyperparameter optimisation of each classifier is done with `RandomizedSearchCV` with three-fold cross-validation on the training data. `RandomizedSearchCV` is used instead of `GridSearchCV` as it samples a set number of combinations from the hyperparameter space, covering it more thoroughly for the same amount of computation and reaching close-to-optimal hyperparameter combinations more quickly (Anirudh S et al., 2024). Macro F1-score is the optimisation criterion to account for the class imbalance.

3.6 Ensemble Voting: Hard and Soft Voting

After optimising individual classifiers, they are ensemble to form an ensemble using scikit-learn's `VotingClassifier` in two configurations. Hard voting takes a vote for each class among the base learners and selects the class with the majority vote. Soft voting takes the average of the predicted probabilities for the classes and selects the class with the highest mean probability. Soft voting would be expected to yield better results than hard voting when the base learners are able to produce highly accurate probability estimates, due to the use of the entire confidence distribution vs a binary vote (Bountakas and Xenakis, 2023). The `VotingClassifier` is set to weight all six base learners equally; no post-fitting tuning of the weights was done, as the goal was to test the ensemble as is, rather than with a tuned weight distribution. Training and testing is conducted on the same splits for both voting schemes.

3.7 Generalisation: Leave-One-Source-Out

A Leave-One-Source-Out (LOSO) cross-validation scheme is followed to evaluate generalisation. The training and test sets are composed of all source datasets except one, which is then used as a test set. Each source is used as the test set once. LOSO is preferred to conventional k-fold cross-validation, where folds are created by random sampling which results in training and test folds having common source characteristics that obscure cross-domain problems. LOSO directly

evaluates the model on a source not included in training, mimicking the real-world scenario where a model trained on one set of emails will be tested on emails from a new source or new campaign (Henry, Stephen and Asuquo, 2026; Mia, Derakhshan and Pritom, 2024). The MeAJOR Zenodo data release preprocessed data includes three different source identifiers (TREC-05, TREC-06, TREC-07), resulting in three LOSO folds. Results are given as the mean macro F1 and standard deviation of the folds.

3.8 Model Interpretability

Logistic Regression is not only used as a classifier, but also the main interpretability tool. After optimisation, the top 20 positive coefficient features (terms most associated with phishing) and the top 20 negative coefficient features (terms most associated with benign) are identified, ranked by coefficient size and examined. This coefficient analysis explicitly confirms the model has learned linguistically sensible discriminative features, and not dataset-specific spurious correlations (Mia, Derakhshan and Pritom, 2024). Note that Logistic Regression coefficients correspond to linear marginal effects on the log-odds, and do not account for feature interactions and non-linearities learned by the ensemble members (e.g. Random Forest). This means that the coefficient analysis offers a clear, but limited view of the decision-making process of the entire pipeline.

4. Evaluation

This section describes quantitative evaluation using four complementary approaches: conventional classification metrics, false positive rate/false negative rate, McNemar's statistical test of significance and leave-one-site-out (LOSO) cross-validation for generalisation. Each model is assessed on the same a 20 percent stratified test set from the 103,340-email data set. We report macro-averaged F1 to prioritise equal performance across the two classes (phishing and non-phishing) independent of their respective numbers, in line with the macro F1 optimisation criterion used during hyperparameter tuning, and in situations where both false negatives and false positives are operationally significant (Kumar S et al., 2025).

4.1 Performance on the Test Set

The accuracy, precision, recall and F1 (all macro-averaged) for the six standalone classifiers and two ensemble methods on the test set . Overall the Soft Voting ensemble performs best in all four measures, with a macro F1 of 97.87 percent, an accuracy of 97.90 percent, a precision of 97.90 percent and a recall of 97.84 percent. Second highest overall performance is that of the Hard Voting ensemble, at 97.56 percent macro F1, which shows that averaging probabilities is superior to voting by majority class. Of the stand-alone classifiers, LinearSVC achieves the highest performance (97.54 percent macro F1), followed by Logistic Regression (97.07 percent), Random Forest (96.85 percent), kNN (95.96 percent) and MNB (94.16 percent). The poorest performing individual classifier is AdaBoost (84.90 percent macro F1).

The 0.33 percentage point macro F1 difference between Soft Vote and LinearSVC is small, as per the ensemble theory principle that such an ensemble of classifiers with different error patterns leads to stable but marginal improvements in already high-performing systems (Bountakas and Xenakis, 2023; Anirudh S et al., 2024). The Soft Vote ensemble's macro F1 of 97.87 percent is comparable to the state-of-the-art HELPHED system's F1 of 0.9942 (Bountakas and Xenakis, 2023); the only improvement left is to add content features to the text-only TF-IDF features used here.

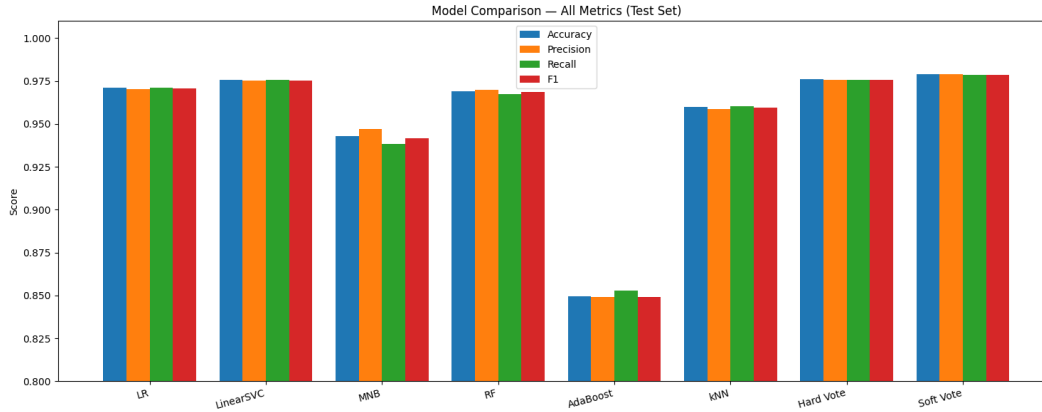


Figure 3: Model Comparison — All Metrics

Model	Accuracy	Precision	Recall	Macro F1
Soft Vote	97.90%	97.90%	97.84%	97.87%
Hard Vote	97.59%	97.56%	97.57%	97.56%
LinearSVC	97.57%	97.52%	97.56%	97.54%
LR	97.10%	97.02%	97.13%	97.07%
RF	96.89%	96.98%	96.75%	96.85%
kNN	95.99%	95.88%	96.05%	95.96%
MNB	94.27%	94.72%	93.84%	94.16%

Table 1: Classification performance on the held-out test set (macro-averaged metrics)

4.2 Analysis of False Positive and False Negative Rate

Unqualified accuracy measures hide important operational failure modes. False negatives (FN) - phishing emails classified as legitimate - are direct security threats, while false positives (FP) - legitimate emails classified as phishing - reduce analyst confidence, and cause operational costs. Figure 4 shows the elements of confusion matrices and the resulting FP and FN rates.

The Soft Voting ensemble has the lowest FP rate (1.66 percent) and the least number of false alarms among all models (Bountakas and Xenakis, 2023). LinearSVC has the lowest FN rate of 2.50 percent, narrowly beating Soft Vote (2.66 percent) on the most critical (to security) metric. This is

a functional trade-off: if the threat of missed phishing is the greater concern, LinearSVC should be preferred; if managing false alarms is of greatest concern, then Soft Vote should be preferred.

AdaBoost has the highest FP rate (17.85 percent) and FN rate (11.59 percent). This is probably due to a mismatch between AdaBoost's weak learners (i.e. decision tree stumps, single-feature threshold classifiers) and the high-dimensional sparse feature set used (i.e. TF-IDF). Decision stumps are not expressive enough to learn informative splits from the bag-of-words representation with 20,000 features, as there is no single feature that is informative enough to split on. This is consistent with the fact that AdaBoost needs more powerful weak learners for text classification. RandomOverSampler did not resolve this issue; future research should investigate AdaBoost with deeper decision stumps and other boosting methods (such as Gradient Boosting) that are based on more expressive base learners. MNB performs well in terms of a low FP rate (2.01 percent) but a high FN rate (10.31 percent): the probability smoothing prevents over-flagging legitimate emails but loses a significant number of phishing emails - a very worrying combination for security-related tasks.

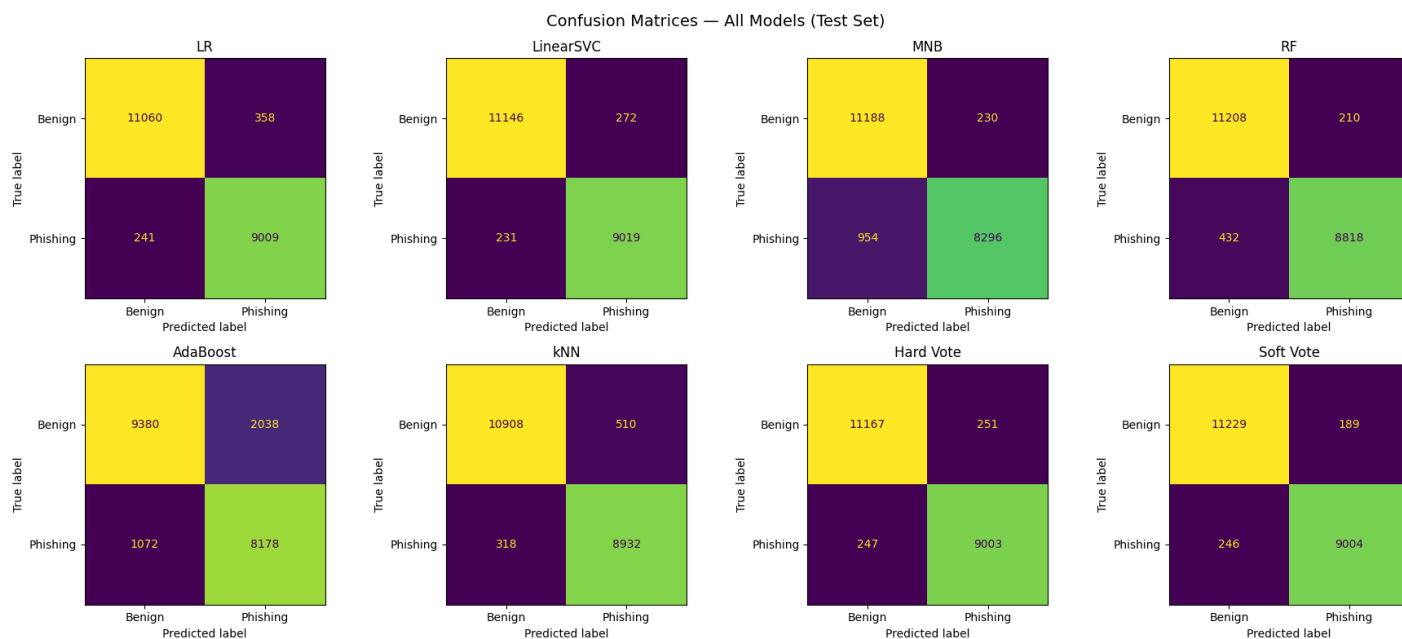


Figure 4: Confusion Matrices — All Models

Model	TN	FP	FN	TP	FP Rate	FN Rate	Model
LR	11,060	358	241	9,009	3.14%	2.61%	LR
LinearSVC	11,146	272	231	9,019	2.38%	2.50%	LinearSVC
MNB	11,188	230	954	8,296	2.01%	10.31%	MNB
RF	11,208	210	432	8,818	1.84%	4.67%	RF
AdaBoost	9,380	2,038	1,072	8,178	17.85%	11.59%	AdaBoost
kNN	10,908	510	318	8,932	4.47%	3.44%	kNN

Table 2: Confusion matrix components and FP/FN rates on the held-out test set

4.3 Statistical Significance: McNemar's Test

We performed McNemar's test on each pair of Soft Vote and another model using the discordants to calculate a continuity-corrected chi-squared statistic $(|b - c| - 1)^2 / (b + c)$. We show the results for all comparisons in Table 3.

Soft Vote is statistically significantly better than all the individual classifiers and Hard Vote ($p < 0.0001$), giving strong evidence that aggregation of models leads to reliable gains in performance. The chi-squared for AdaBoost is the largest ($\chi^2 = 2398.62$), due to the large performance difference. It's worth noting, though, that statistical significance is not necessarily equivalent to practical significance. Soft Vote is statistically better than LinearSVC ($p < 0.0001$, $\chi^2 = 17.54$), but the 0.33 percentage point gain in F1 is not significant for most applications. In security applications where FN is the main concern, like medical diagnosis, LinearSVC's lower FN rate of 2.50 percent is preferred over Soft Vote's statistically better macro F1. Here, statistical significance indicates the gain is not spurious, but doesn't otherwise decide which model to deploy.

Model vs Soft Vote	b	c	Chi ²	p-value	Significant?
LR	233	69	87.98	< 0.0001	YES
LinearSVC	162	94	17.54	< 0.0001	YES
MNB	833	84	610.15	< 0.0001	YES
RF	334	127	92.05	< 0.0001	YES
AdaBoost	2,828	153	2398.62	< 0.0001	YES
kNN	568	175	206.82	< 0.0001	YES

Table 3: McNemar's test — Soft Voting ensemble vs each model (continuity-corrected)

4.4 ROC-AUC and Precision-Recall

ROC-AUC analysis confirms the ranking of the models by macro F1 in Section 4.1. Soft Voting ensemble has the highest AUC of 0.9978, followed by LinearSVC (0.9964), Random Forest (0.9962) and Logistic Regression (0.9954). MNB is 0.9913, kNN 0.9839, and the lowest is 0.9326 for AdaBoost - consistent with its poor macro F1 score and suggesting its failure is not due to a poor choice of threshold, but rather a weakness in discrimination.

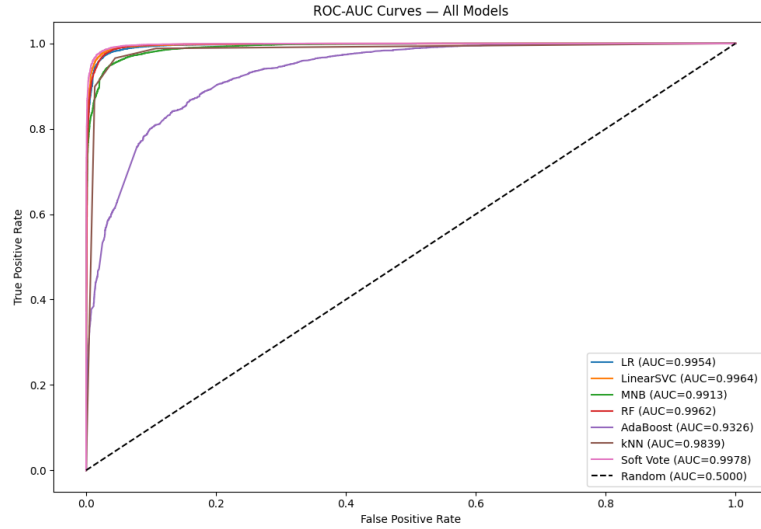


Figure 5: ROC-AUC Curves

The Precision-Recall curve analysis for Soft Vote achieves an average precision (AP) of 0.9972, confirming that it has high precision across the range of recalls - a desirable characteristic of imbalanced classification as recall is the critical operational axis. LinearSVC has $AP = 0.9954$, in line with its FN performance. Soft Vote's Precision-Recall threshold analysis finds a deployable operating point at $\text{recall} \geq 0.98$ that can be tuned for high-sensitivity security deployments that need to capture close to all phishing emails at the expense of a small increase in false alarms, as elaborated in Section 6.3 (Bountakas and Xenakis, 2023).

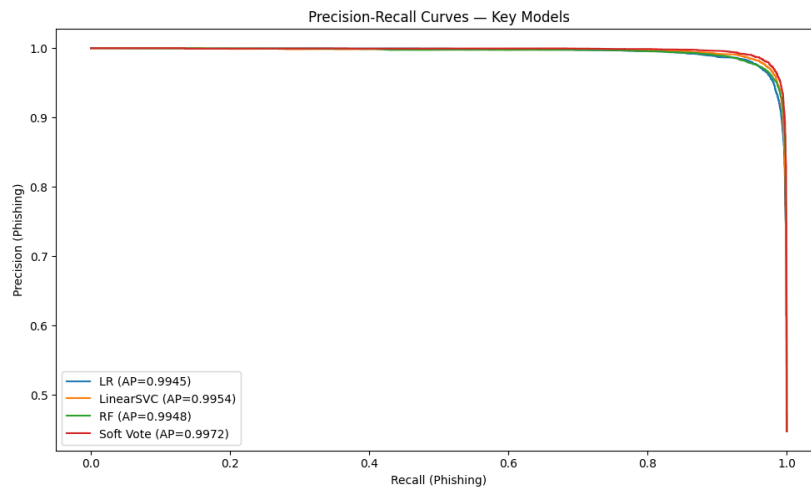


Figure 5: Precision-Recall Curves

4.5 Generalisation: Leave-One-Source-Out (LOSO)

LOSO macro F1, for the three source folds held out, along with means and standard deviations are shown in Table 5. All models' LOSO F1 values are lower than those of in-distribution test, confirming that cross-source deployment is a more complex generalisation task than in-distribution testing (Henry, Stephen and Asuquo, 2026; Mia, Derakhshan and Pritom, 2024).

Soft Vote achieves the second-highest mean LOSO F1 at $84.94\% \pm 5.03\%$, behind MNB at $85.98\% \pm 6.68\%$. Moreover, Soft Vote's standard deviation is lower, showing more robust cross-source behaviour - the most important deployment characteristic. The 12.93% F1 drop between Soft Vote's in-distribution and LOSO means measures the cost of generalisation for cross-source deployment, highlighting the importance of multi-source training and cross-source evaluation.

The superior mean LOSO F1 of MNB despite the lower in-distribution performance is the most surprising. Its probabilistic smoothing is an implicit regulariser that prevents overfitting to source-specific words, whereas the Naive Bayes conditional independence assumption may paradoxically enhance cross-source generalisation by treating features as interchangeable pieces of evidence, rather than learning source-specific co-occurrences. This is consistent with MNB's highest LOSO F1 on TREC-06 (91.85%) that has the most unique ratio of phishing to benign examples.

One important thing is that with only three source folds, mean and standard deviation estimates are statistically non-robust and easily influenced by single-fold results. LOSO results should be interpreted with caution; with five or more folds, these statistics would be stable. RF has the lowest standard deviation (4.07%) but only a moderate mean F1 (80.36%) while kNN (72.36%) and AdaBoost (68.91%) perform poorly in cross-domain settings.

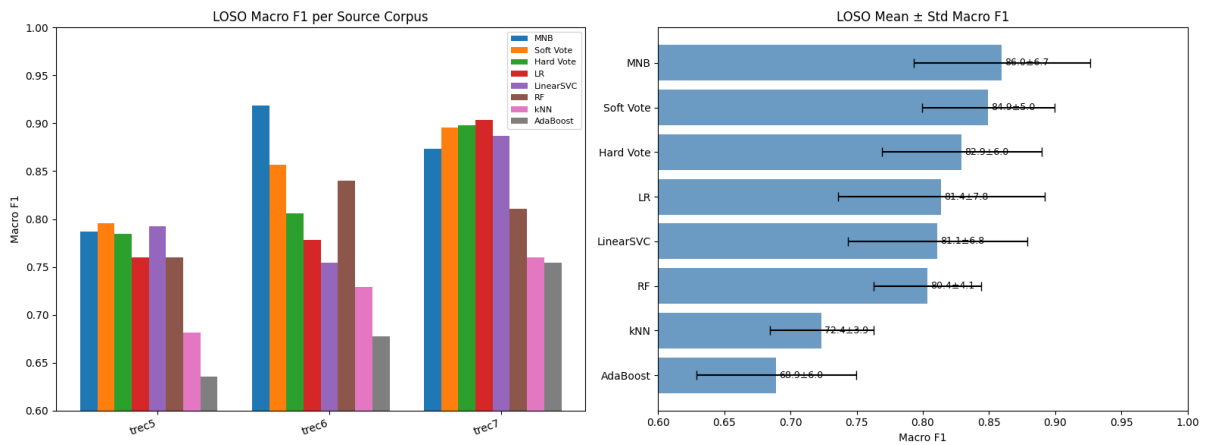


Figure 6: LOSO F1 per Source + Mean ± Std

5. Model Interpretability

After hyperparameter tuning, the TF-IDF coefficient weights of the Logistic Regression predicted values were extracted and sorted. A positive coefficient will predict a phishing email, a negative

coefficient the other way. The top 20 of each are shown below. It's worth noting that Logistic Regression coefficients are linear marginal effects on the log-odds and don't capture the interactions and non-linear effects that other classifiers like Random Forest might capture; the analysis therefore offers a clear but necessarily simplistic insight into the discriminative process of the pipeline (Mia, Derakhshan and Pritom, 2024).

5.1 Top Phishing Indicator Features

"url" (coefficient +5.3645) is the top predictor of phishing and this is unsurprising: URLs are a commonly recognised feature of phishing attacks (Mendes, Maia and Praça, 2025). Perhaps more interesting are some of the top features that are not discussed in other phishing literature. "emoji" (+3.9876) is the seventh most discriminative feature - ahead of "click" (+3.2715) and "medication" (+3.5552) - and it is likely to be a strong phishing predictor because of the use of HTML-encoded emoji characters in phishing subject lines and email body as a way to grab attention and evade keyword-based anti-spam filters, a tactic increasingly used in recent phishing emails. "printer" (+2.9106) and "watch" (+3.2354) are unexpected phishing features: "watch" could be related to luxury goods impersonation scams while "printer" is likely related to technical support phishing templates impersonating printer manufacturers. "quality" (+3.9142) likely corresponds to commercial spam that co-exists with phishing in the TREC corpus, where junk email campaigns are using persuasion tactics related to quality.

"loan" (+3.9812), "mortgage" (+2.9152), "medication" (+3.5552), "med" (+3.9635), and "price" (+3.8430) collectively reflect financial and pharmaceutical spam with language that has commonalities with phishing that attempts to steal credentials. "offer" (+4.0772) and "save" (+4.0503) are urgency persuasion language typical of social engineering schemes (Bountakas and Xenakis, 2023). "adf" (+3.0953) is probably an attribute of an advertisement framework in phishing email HTML - a data source artefact in TREC emails. Its inclusion at position 17 demonstrates a key point: some high coefficient features are specific to the dataset and reflect phishing artefacts and not global phishing indicator features, as identified by Mia, Derakhshan and Pritom (2024) using SHAP.

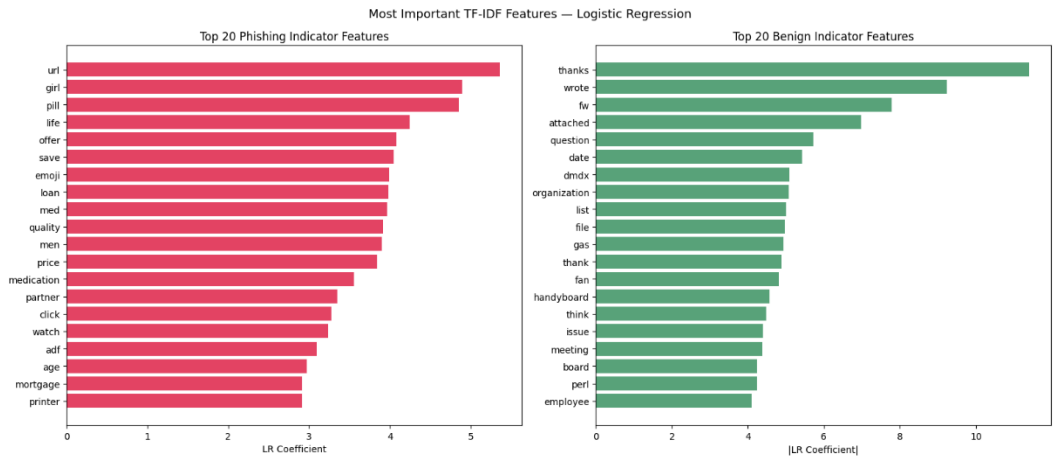


Figure 5: TF-IDF Feature Importance

5.2 Top Benign Indicator Features

"thanks" (coefficient -11.4075) is the top benign indicator, more than twice the magnitude of the top phishing indicator. This imbalance - benign features are more definitively distinctive than phishing features - mirrors the very predictable polite social style of professional email that is seldom mimicked in phishing emails. "wrote" (-9.2306) is an element of email reply chains ("On [date], [person] wrote:"), and is a strong indicator of legitimate email. "fw" (-7.7841) suggests forwarded business emails and "attached" (-6.9765) refers to legitimate attachment-sharing practices in business email typical of the Enron Corpus component of MeAJOR.

"gas" (-4.9402) is the 11th benign indicator and is worth explaining: it is likely an indicator of the Enron Corpus component of MeAJOR, which involves a large number of internal Enron corporate emails about energy trading and gas markets. This is highly domain-specific and would not extend to organisations not in the energy industry, and is an example of the explainability problem that source-specific words create high-weight features which have non-transferable discriminative information. Likewise, "dmdx" (-5.0877), "handyboard" (-4.5671), "perl" (-4.2435), and "fan" (-4.8226) are technical terms from the emails of science and engineering mailing list emails in the TREC corpus. Terms of workplace organisation - "meeting" (-4.3847), "organisation" (-5.0855), "employee" (-4.0981), "issue" (-4.3895) - and terms of collaboration - "question" (-5.7263), "think" (-4.4833) - reflect the language of legitimate business emails which is absent from phishing emails, as desired for explainability as identified by Anirudh S et al. (2024) and Bountakas and Xenakis (2023).

6. Conclusions and Future Work

6.1 Conclusion

In this project, we have built an ML pipeline to identify phishing emails with a corpus of 108,685 samples, in-distribution and LOSO cross-validation were considered. Our key observations are: The best in-distribution performance was observed with the Soft Vote ensemble model (macro F1 = 97.87%, FPR = 1.66%); it is statistically significant (according to McNemar, p-value < 0.0001). The generalisation difference that was reached through LOSO was 12.93% and yet, the Soft Vote model is the most deployable. The tradeoff of the minimum number of false positives (in case of Soft Vote) and the number of false negatives (in LinearSVC model) is valid. Further, Linguistic feature learning has been established in the Logistic Regression model as well as source related bias which hinders generalisation to cross sources. Hence, this result is an indication of the usefulness of TF-IDF ensembles to obtain the same state-of-the-art in-distribution outcomes with resource-limited settings.

6.2 Limitations

Three limitations apply. First, the pipeline only uses TF-IDF features; no URLs, headers or attachment-based signals are included - it likely won't outperform hybrid models on obfuscated phishing. Second, hyperparameters for LOSO were selected on the whole dataset, so data from the sources that weren't used influenced the model choice; cross-domain accuracy might be lower when the hyperparameters are returned for each fold. Third, kNN is not scalable, and is used only for experimentation.

6.3 Future Work

Three extensions are prioritized. Most simply, a hybrid feature matrix can be constructed by appending TF-IDF with URL lexical features and email header features using `scipy.sparse.hstack` - an easily implemented, proven improvement to the model. Second, the contextual semantic modelling of transformer embeddings (BERT, RoBERTa) is impossible with TF-IDF, and the LOSO framework created here is an excellent test bed. Finally, classification threshold optimisation using precision-recall curves and cost-based models should better adjust model error characteristics - particularly for the LinearSVC versus Soft Vote trade-off - to specific security needs.

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